AnnaSetu AI — Platform Field Guide

“Share More. Waste Less. Feed Hope.” | Complete Plain-English Explanation of Every Screen & Field

Platform: AnnaSetu AI Surplus Food Coordination | **Focus:** UN Sustainable Development Goals: SDG 2 (Zero Hunger) & SDG 12 (Responsible Consumption)
Scope: Every form field, button, status code, calculation, and safety check explained simply for donors, NGOs, volunteers, and evaluators.

1. User Roles & Account Capabilities

AnnaSetu AI enforces strict Role-Based Access Control (RBAC). Every user interacts with a tailored interface designed around their real-world responsibilities:

Role	Who Uses It	What They Can Do & Key Restrictions
Food Donor	Restaurants, banquet halls, college hostel kitchens, catering services, corporate cafeterias.	Post surplus food batches; use AI categorization; view matched NGOs; approve reservation requests; monitor courier pickup. Cannot view other donors' listings.
Recipient Organization	Verified NGOs, registered community kitchens, homeless shelters, child/elder relief homes.	Search food batches via AI natural language; reserve meals within daily capacity. **Security Gate:** Must be admin-verified with 12A/80G/Darpan before reserving.
Volunteer Courier	Good Samaritan volunteer drivers, community youth, vehicle owners with transport crates.	Browse unassigned pickup runs across Bengaluru; claim runs; confirm kitchen collection; confirm shelter handover. Cannot view private donor finances.
Administrator	Central compliance team, food safety auditors, platform moderators.	Approve/reject NGO accreditation; investigate food safety/spoilage alerts; moderate listings; inspect immutable tamper-evident audit logs. Seeded only via secure config.

2. Registration & Account Setup Fields (`/register`)

When an organization or volunteer registers, each field serves a specific security and operational purpose:

Form Field	Easy Plain-English Meaning	Why It Matters & Rules
Full Name	Contact person's full official name (e.g., Ramesh Kumar).	Used on delivery invoices and handover verification.
Email Address	Login username and notification email destination.	Must be unique; used to send critical pickup alerts.
Password	Secure password (min 6 characters).	Hashed using Argon2 algorithm; never stored in plaintext.
Phone Number	Direct mobile number for coordinating pickup handovers.	Shared strictly between donor, assigned volunteer, and recipient.

I am registering as	Account role dropdown (Food Donor, Recipient NGO, Volunteer).	Determines dashboard navigation, allowed actions, and APIs.
Organization Name	Legal name of hotel, restaurant, NGO, or shelter.	Required for Donors and Recipients; displayed on match badges.
Organization Type	Type of establishment (e.g. Restaurant, Hotel, NGO, Shelter).	Helps the AI matching engine recommend suitable meal types.
Registration / License No.	FSSAI license number (donor) or 12A/80G/Darpan ID (NGO).	Crucial trust verification field for compliance auditing.
Address & City	Physical street address where kitchen or shelter is located.	Used to calculate road distance and dispatch volunteers.
Daily Capacity (Servings)	Max meals the NGO shelter can safely distribute per day.	**Prevents Overbooking:** An NGO with capacity 100 cannot request 500 meals.
Service Radius (km)	Maximum distance the NGO or volunteer can travel (default 15 km).	Limits matches to food within reasonable travel time.

3. Donor Surplus Food Listing Form (`/donor/listings/new`)

This is the core donation entry form. Food safety rules are strictly enforced over raw AI output:

Field Name	Explanation & Supported Values	Food Safety & System Impact
Food Title	Clear name of food (e.g. 'Fresh Paneer Butter Masala & Jeera Rice').	Must be descriptive. The AI uses this to detect allergens.
Food Category	Dropdown: Prepared Meals, Bakery, Produce, Packaged Food, Dairy, Other.	Determines allowable transit time and storage protocols.
Quantity & Unit	Numeric amount (e.g. 15.0) and unit (kg, liters, packets, portions).	Must be greater than 0. Tracked atomically in inventory.
Estimated Servings	Number of individual people this batch can feed (e.g. 45 meals).	Matched against NGO recipient shelter capacity to prevent waste.
Preparation Date & Time	Exact date and time when the meal was prepared/cooked in kitchen.	Used to calculate freshness window. Cannot be in the future.
Expiry / Best-Before	Time cutoff after which food must NOT be consumed.	Listings expire automatically in the database past this date.
Storage Conditions	**hot_holding** (≥60°C) **refrigerated** (≤5°C) **ambient** (room temp) **frozen** (≤-18°C)	**FSSAI Mandate:** Hot foods must stay hot, chilled must stay cold during transit. Couriers check this on arrival.
Known Allergens	Text tag of allergens present (e.g. 'Dairy, Gluten, Nuts').	Protects beneficiaries with severe dietary allergies.
Dietary Attributes	Dropdown: vegetarian, vegan, non_vegetarian, jain, halal.	Allows shelters with strict dietary guidelines to filter accurately.
Pickup Street Address	Exact pickup location (e.g., 'Grand Palace Hotel, MG Road Kitchen Bay 2').	Coordinates are used by the Haversine formula for proximity.

Pickup Available From	Earliest time food is packaged and ready for collection.	Ensures volunteers don't arrive before food is packed.
Pickup Deadline	Latest time courier can pick up food before kitchen closes.	**Strict Rule:** Must be in the future. Urgent if <3 hours away.
Contact Name & Phone	Duty chef or banquet manager on site (e.g. 'Chef Ramesh', '9845012345').	Direct communication link visible ONLY to matched participants.
Food Safety Declaration	Mandatory confirmation checkbox: *'I confirm this food has been prepared hygienically and is safe for consumption.'*	**Legal Requirement:** Listing CANNOT be submitted without this confirmation. Preserves audit accountability.
AI Smart Suggestions	Clickable co-pilot button beside title/description.	Uses Gemini AI (or rule fallback) to auto-fill category, detect hidden allergens, and improve packaging notes.

4. Intelligent Proximity Matching & Scoring (0–100)

AnnaSetu AI does not use a black-box opaque algorithm. Every recommendation displays transparent breakdown factors:

Scoring Factor	Weight / Points	What It Evaluates in Plain English
Geodesic Distance (Haversine)	Up to 40 points	Proximity between donor kitchen and NGO shelter. Closer = higher score (e.g. <5 km gets max points). Eliminates long transit times that risk spoilage.
Shelter Daily Capacity Fit	Up to 30 points	Checks if NGO's daily capacity matches the batch size. Prevents sending 200 meals to a 30-bed shelter or 10 meals to a 500-person camp.
Schedule Urgency Window	Up to 20 points	Rewards rapid action for food expiring soon. If deadline is within 3 hours, matching priority escalates.
Food Category / Dietary Match	Up to 10 points	Ensures ready-to-eat cooked meals go to immediate dining shelters, while bulk raw produce goes to kitchens with cooking facilities.

5. Volunteer Logistics & Status Lifecycle

Every food donation progresses through an explicit, auditable state machine. Couriers progress the mission using single-click buttons:

Status Code	Meaning & Who Triggers It	What Happens Under the Hood
AVAILABLE	Food posted by donor; visible in the catalog.	Unreserved inventory. Open for verified NGOs to browse and request.
REQUESTED	NGO submitted a claim for this food batch.	Row-level lock holds inventory. Donor receives notification to approve.
ACCEPTED	Donor approved the NGO reservation request.	Creates an unassigned pickup task in the volunteer dispatch queue.

ASSIGNED	Volunteer claims or is assigned to the run.	Volunteer reviews donor kitchen address, phone, and pickup deadline.
PICKED_UP	Volunteer collects food from donor kitchen.	Volunteer enters thermal condition notes; recipient shelter notified food is in transit.
COMPLETED	Volunteer delivers food safely to recipient shelter.	Shelter supervisor signs off; impact metrics automatically increment in database.
CANCELLED / FLAGGED	Cancelled by donor or flagged by admin.	Removed from active matching; logged in audit ledger with reason.

6. Administrator Governance & Compliance (`/admin`)

The Admin Command Center ensures platform safety, trust, and accountability:

Admin Section / Field	Plain-English Purpose	Action Performed by Auditor
Accreditation Queue (`/admin/verifications`)	Inspect NGO submitted 12A/80G documents or NGO Darpan certificates.	Click **Approve & Verify** to unlock food requests for that NGO, or **Reject** with feedback notes.
Safety Incidents (`/admin/safety-reports`)	Investigate reports of spoilage, late pickups, or cold chain breaks.	Assign investigation status (`investigating`, `resolved`, `dismissed`) and flag non-compliant donors.
Catalog Moderation (`/admin/listings`)	Platform-wide view of all active, completed, and expired listings.	Cancel or flag listings that violate hygiene regulations or contain prohibited foods.
User Accounts (`/admin/users`)	Directory of all registered donors, NGOs, and volunteers.	Toggle active/suspended status to block compromised accounts (admin self-deactivation blocked).
Immutable Audit Ledger (`/admin/audit-logs`)	Append-only log of every login, status change, verification, and match.	Inspect timestamp, user email, action name, and JSON details for compliance auditing.

7. Impact Dashboard Metrics (`/impact`)

All metrics displayed on AnnaSetu AI are computed live using database queries with zero fabricated numbers:

Metric Displayed	Database Calculation Formula	Real-World Sustainable Impact
Total Servings Redistributed	SUM(estimated_servings) for listings with status = 'COMPLETED'	Directly advances **SDG 2 (Zero Hunger)** by tracking individual plates provided to vulnerable citizens.
Quantity Diverted (kg)	SUM(quantity) converted to kg for completed donations	Advances **SDG 12** by measuring edible organic matter prevented from rotting in municipal landfills.
Pickup Completion Rate (%)	(Completed Pickups / Total Created Pickups) * 100	Measures volunteer dispatch reliability and logistical effectiveness across the city.
Category Breakdown	Grouped counts by Prepared Meals, Bakery, Produce, Packaged Food	Reveals whether surplus is primarily ready-to-eat banquets or fresh raw market surplus.

8. AnnaSetu AI Assistant Floating Widget

Located at the bottom-right corner of every page, the AI Assistant provides interactive, role-scoped guidance:

• **Role-Scoped Answers:** A donor asking about their listings gets kitchen-specific advice; an NGO asking about eligibility receives step-by-step accreditation instructions.
• **Natural Language Search:** Converts phrases like *'Need 50 vegetarian meals for dinner in Indiranagar'* into structured database filters.
• **Resilient Fallback:** If Gemini API quota or network connectivity is unavailable, the assistant continues functioning using high-accuracy deterministic rule matching.